

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

November 2013

IT403 Data Compression

Date: 14.11.2013, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Do as directed.

[07]

1. Define: 1st Order Entropy H of a Source Alphabet.
2. Define: Self Information of a symbol.
3. Define: Prefix Code.
4. Check for Uniquely Decodable property. $\alpha = \{0,01,11,011,101,10100\}$
5. Fill in the Blank: When source alphabet contains less number of symbols with similar probabilities _____ is generally preferred for higher compression ratio. (Arithmetic Coding, Extended Huffman Coding)
6. Fill in the Blank: Data Compression is an _____ and _____ of representing data in _____ form.
7. A source is containing 2^N equi-probable symbols. If, initially N bit fixed length code is assigned to represent source symbols, then no other coding algorithm can be found which can compress the data. (True/False).

Q – 2 Answer following questions.

[14]

- (a) Is Data Compression=Modeling +Coding? Justify using suitable examples. 5
- (b) Explain performance measurement parameters for Data Compression algorithms. 5
- (c) Using probabilities $P(A)= 0.51$, $P(B)=0.22$, $P(C)=0.17$, $P(D)=0.07$, and $P(E)=0.03$, 4
Find out shanon fano code, source entropy, average length and compressoin ratio.

OR

Q – 2 Answer following questions.

[14]

- (a) If probabilities of all the symbols of an alphabet are in order of 2^{-n} (where, n is integer), then Huffman Coding performs best. (True /False). Justify your answer by providing suitable example. 5

- (b) For a binary source with probabilities $P(A) = 0.99$, $P(B) = 0.01$, design a Huffman code for the source obtained by blocking m symbols together, where $m = 1, 2$. Plot the average length vs. m graph. Comment on your results. 5
- (c) Explain Physical model with suitable examples in context of Data compression. 4

Q – 3 Answer following questions. [14]

- (a) When arithmetic coding is preferred over Extended Huffman coding? Why? 3
- (b) Differentiate following: 1. Lossy Compression vs. Lossless Compression 4
2. Statistical vs. Dictionary based compression
- (c) Consider a source containing 10 distinct symbols [0-9]. Encode given sequence of symbols using Adaptive Huffman algorithm. 7
Symbol Sequence: 0, 1, 1, 2, 3, 5, 8.

Q – 3 Answer following questions. [14]

- (a) Explain Sibling property in Adaptive Huffman Coding. 3
- (b) Explain process of generating Non-Binary Huffman Code. 4
- (c) Given source with probabilities of symbols as $P(A)=0.4$, $P(B)=0.3$, $P(C)=0.2$, $P(D)=0.1$. Perform decoding of string using arithmetic coding tag given as below. 7
Tag: 0.23758964

SECTION – II

Q – 4 Check the correctness of following statements and Justify the choice within one statement. [07]

1. Higher the resolution of an image leads to High Quality of an image.(T/F)
2. In JPEG compression, Discrete Cosine Transform is a lossy step.(T/F)
3. Quantization is a process of decreasing redundancy.(T/F)
4. Image/Audio/Video is compressed using lossy techniques because they contains highly correlated data. (T/F)
5. Low frequencies in frequency domain of an image represents smooth area of an image in spatial domain.(T/F)
6. In JPEG compression, different quantization tables are used for controlling quality of an image.(T/F)
7. In LZW decoding process, it is possible that an index in dictionary can be referred for decoding, when entry of that index is partially done.(T/F)

Q - 5 Answer following questions.

[14]

- (a) Encode following sequence using Digram Coding technique with 3-Bit index field. 3

Sequence: RATATATATRARATRAATATA

- (b) How to get number of bits for LZ77 Triple? What will be the effect of changing window size and search buffer size on performance of Compression? 5

- (c) Decode following LZ77 Triples and retrieve original string. Consider Source Alphabet={a, b, c, d}, Search Buffer Size =5 characters ,Window Size=10 characters. 6

LZ77 Triples: <0,0,b>, <0,0,a>, <0,0,d>, <2,5,b>, <2,1,a>, <4,3,b>, <3,8,c>

OR

Q - 5 Answer following questions.

[14]

- (a) Compare & contrast LZ78 and LZW algorithms. 3

- (b) Encode following string using of LZ78 algorithm. 5

String: Bed Spreaders spread spreads on beds<END>

- (c) Decoding following index sequence using LZW algorithm. Consider initial dictionary given as below. 6

Index	0	1	2	3	4	5	6
Entry	<END>	B	E	N	O	R	T

Index sequence: 6, 4, 1, 2, 4, 5, 3, 4, 6, 7, 9, 11, 16, 10, 12, 14, 19, 0

Q - 6 Answer following questions.(Any Two)

[14]

- (a) Explain JPEG Image Compression Process with block diagram. 7
- (b) Sampling and quantization of an audio signal. Discuss various parameters which affect the quality and size of digital audio. 7

- (c) What is the significance of I , B and P Frames in MPEG Video compression? Explain Spatial and temporal redundancies in videos. Find the size of Full RGB video clip of 24 second duration with 320x240 resolution @ 25FPS. 7

- (d) Write short note on following: 7

1. Discrete Cosine Transform in Image Compression
2. Companding Codec
